AR-Quivers of Finite-Dimensional Algebras via GAP

Quick Start

Launch the container

> **Important**
>
> **A VPN is required to access the binder when you are in China.**

Click the badge to launch automatically: launch binder, or open one of the following links:

- Stable version: https://mybinder.org/v2/gh/zhangchenchengSJTU/GAP_AR_Quiver/main
- Testing version: https://mybinder.org/v2/gh/zhangchenchengSJTU/GAP_AR_Quiver/test

Manual launch: open Binder and select the `GitHub` option.

1. In the `GitHub repository name or URL` field, paste `https://github.com/zhangchenchengSJTU/GAP_AR_Quiver`.
2. Click `launch`. To use the testing version, set `Git ref (branch, tag, or commit)` to `test`.
3. Wait for the environment to build. Binder will redirect to the Jupyter Notebook page.

Enter Jupyter Notebook

After the environment loads, the browser address will resemble `https://hub.bids.mybinder.org/user/zhangchenchengsjtu-gap_ar_quiver-???????/tree`. The root directory contains:

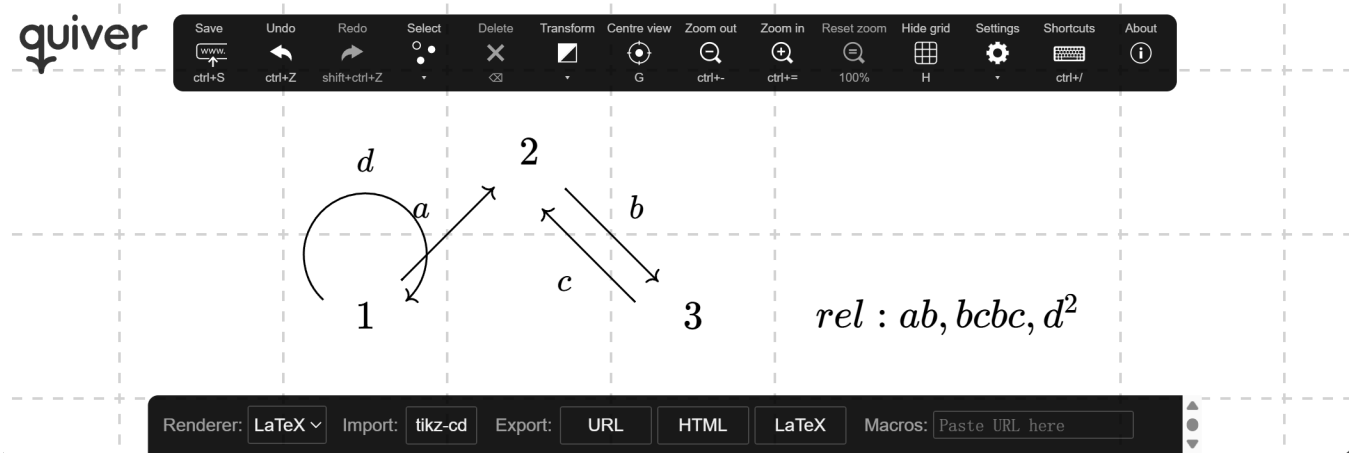
- `ARquiver`: working directory for AR-quiver computation. It contains:
  - `source`: source code directory.
  - `run.ipynb`: notebook for running computation and rendering.
  - `example.txt`, `yourfile.txt`, and similar files: input files containing quiver data.
- `Dockerfile`: environment specification for developers.
- `readme.md`, `readme.html`: documentation.

Draw the path algebra

Use https://q.uiver.app to draw a quiver with relations, following these conventions:

- vertices are positive integers;
- arrows are single Latin or Greek letters in $\LaTeX$ format, e.g. `a` or `\alpha`;
- relations are entered in an empty grid cell in the form `rel: ...`.

Example:



Click `LaTeX` at the bottom of the `q.uiver` page and copy the generated `tikzcd` source code. Create a new file `yourfile.txt` in the `ARquiver` directory, paste the code into it, and save.

Draw the AR-quiver

Open `run.ipynb` and run the following cells in order:

```
# yourfile.txt -> yourfile.log
%run source/compute_all.py
```

```
# yourfile.log -> yourfile.html
%run source/render_all.py
```

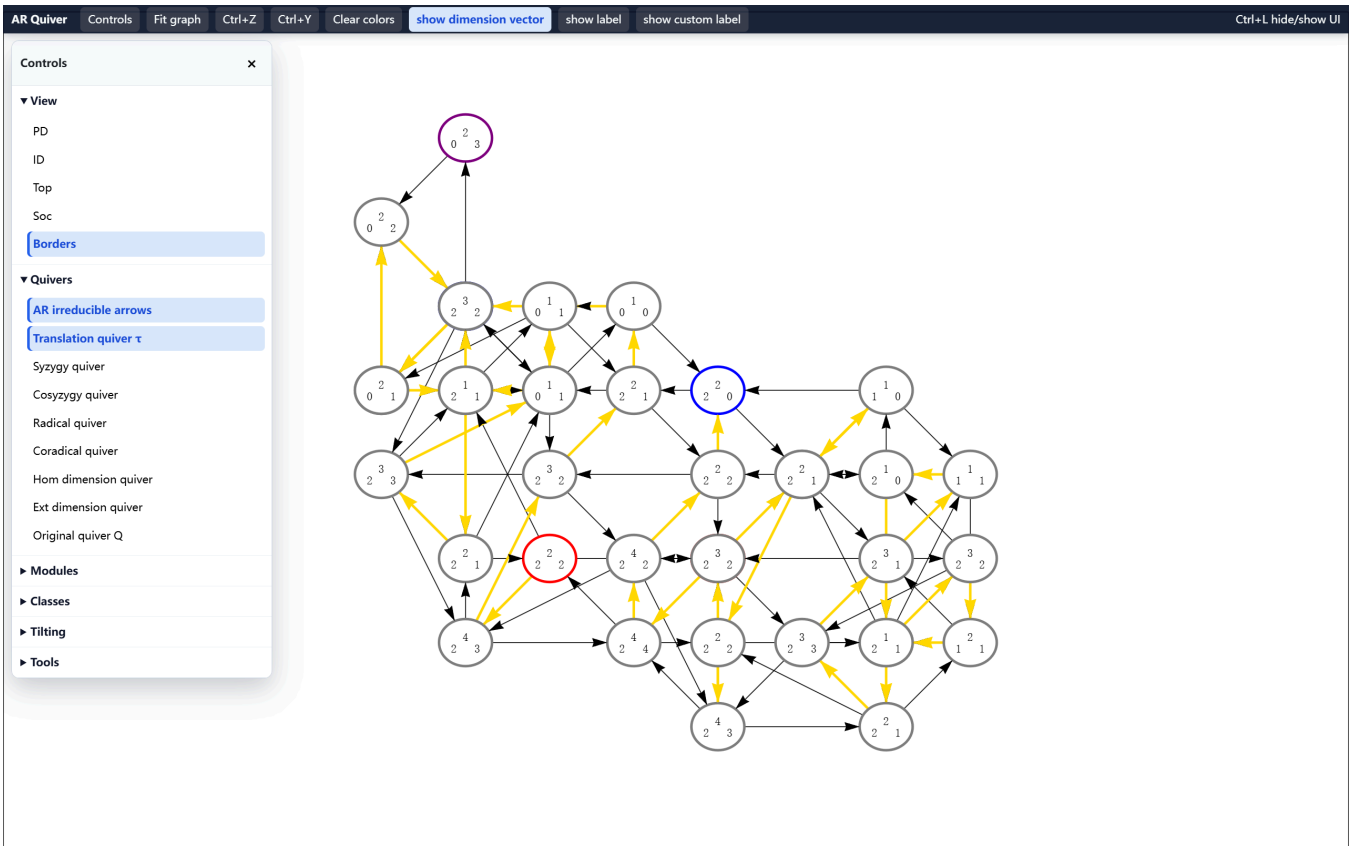
The `ARquiver` directory will then contain:

- `yourfile.log`: algebra computation log;
- `yourfile.html`: interactive AR-quiver canvas.

The `html` file can be downloaded directly:

Arrange the AR-quiver into a usual form

Open `yourfile.html` to view the interactive AR-quiver.



Basic controls:

- Each ellipse represents an indecomposable module; its dimension vector is arranged according to vertex positions in the `tikzcd` input. Some ellipses may overlap.
- Purple border: projective-injective. Red border: projective non-injective. Blue border: injective non-projective.
- `Quivers > AR irreducible arrows`: show/hide irreducible morphisms (black arrows in day mode, white arrows in night mode).
- `Quivers > Translation quiver τ`: show/hide the AR translation $\tau = D\mathrm{Tr}$ (golden arrows).
- `View > Borders`: show/hide vertex borders.
- `History`: browse, replay, undo, and redo recorded UI actions.

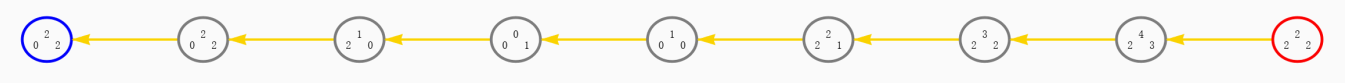
The main task is to arrange the AR-quiver into a readable standard form.

> **Tip**
>
> The τ-orbits are disjoint. Projective-injective objects belong to no τ-orbit; every other indecomposable belongs to exactly one. Each τ-orbit is of one of the following types:
>
> 1. a path from an injective to a projective object;
> 2. a cycle containing no projective or injective object.

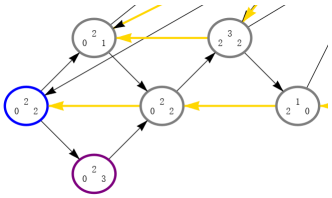
Start by identifying a projective object with a long τ-orbit, e.g., $_0{}^2{}_2$. If vertices overlap, refresh the page to obtain a cleaner initial layout. Hide `AR irreducible arrows` and place the τ-arrow $_0{}^2{}_2 \leftarrow {}_0{}^2{}_2$ horizontally in an empty region.

> **Tip**
>
> **Long-press a golden arrow** to align the subsequent arrow (ensure no black arrows are hidden behind it).

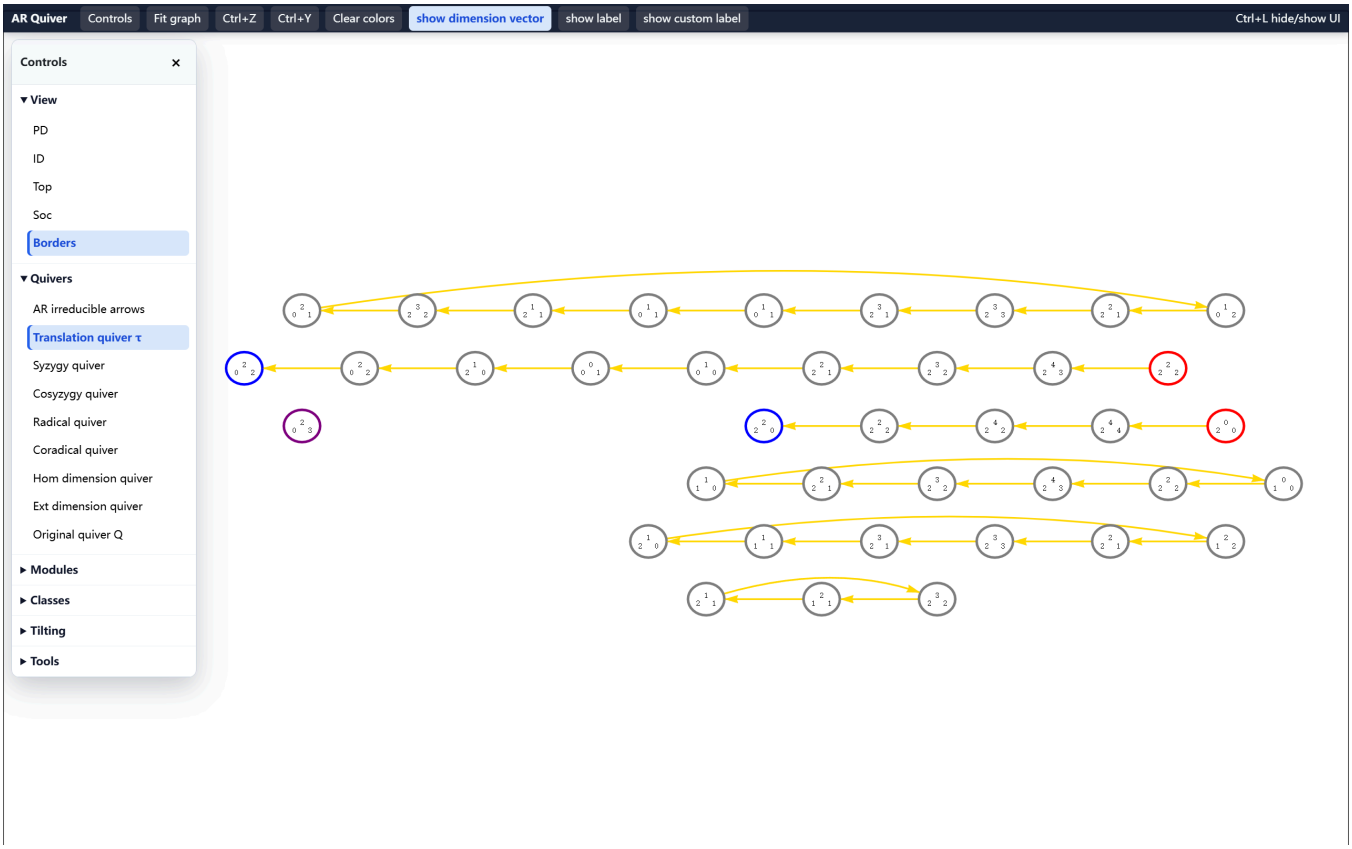
This gives:



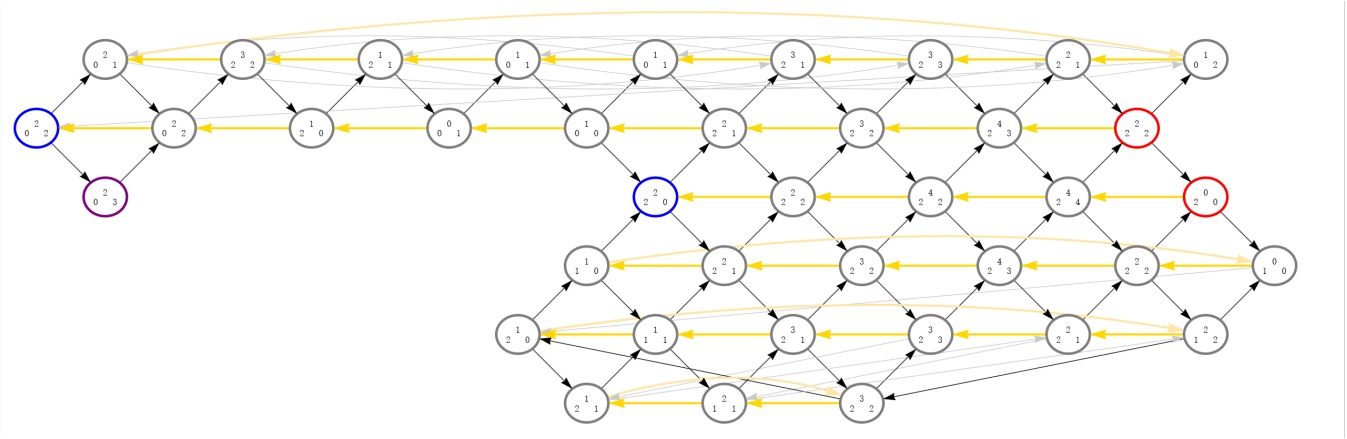
Enable `AR irreducible arrows` and locate the almost split sequence $_0{}^2{}_2 \rightarrowtail \bigoplus M_i \twoheadrightarrow {}_0{}^2{}_2$.



Hide `AR irreducible arrows` again and long-press the golden arrows to complete the alignment. For cycles, select an edge and use `↑` / `↓` to adjust its curvature. After these operations, the τ-orbits are obtained:



Finally, hide `AR irreducible arrows` and adjust the curvature of any black arrows previously hidden behind golden ones. Double-click an edge to toggle between dark and light colours. The final AR-quiver is:



> **Tip**
>
> Press `Ctrl + L` for full screen and enjoy this quiver.

Features

The generated `filename.html` is an interactive workspace for studying the AR-quiver and homological data of the algebra defined by `filename.txt`. The page consists of a main canvas, a top navigation bar, and a sidebar toggled by `Controls`. The sections below follow the interface order: canvas and navigation bar first, then each sidebar section from top to bottom.

Recent additions include the history panel, `Clear canvas`, day/night mode, a direct top-bar `Modules matrix` button, resizable/toggleable floating tool windows, sortable side-drawer lists, module-matrix and class-inspection tools, traversal tools, expanded GAP-code snippets, display-code import/export improvements, and improved night-mode rendering for controls, lists, legends, the original quiver, and module-matrix views.

Basic canvas interaction

Each ellipse represents an indecomposable module. The label is the dimension vector, arranged according to the input quiver shape when available. The internal node number is the identifier used throughout `filename.log`.

- Drag a vertex to reposition it.
- Drag the background to pan.
- Scroll to zoom.
- Hover over a vertex to display its node number.
- Double-click an edge to toggle dark/light colour.
- Select an edge and press `↑` / `↓` to adjust its curvature.

These operations affect only the visualisation and do not recompute the algebra.

Top navigation bar

`Controls`

Opens or hides the sidebar. The sidebar contains all overlays, lists, GAP snippets, and tools, divided into `View`, `Quivers`, `Modules`, `Classes`, `Tilting`, `GAP codes`, and `Tools`.

`History`

Opens a draggable history panel. The panel records recent layout, colouring, class-inspection, and matrix actions where applicable. Use `Back`, `Forward`, or `replay` to revisit a previous state. This replaces the old top-bar `Ctrl+Z` / `Ctrl+Y` buttons while preserving keyboard-oriented workflow through the history panel.

`Fit graph`

Recentres and rescales the canvas to fit the visible graph in the window.

`Clear canvas`

Resets the workspace to the initial visual state without moving vertices. It closes temporary drawers and tool windows, clears list selections and temporary colourings, restores the default label mode, and leaves only the initial graph layers enabled: vertex borders, AR irreducible arrows, and the translation quiver.

`Modules matrix`

Opens the module-matrix viewer directly from the top bar. The same tool is also available from `Controls > Tools > Module matrix`.

`show dimension vector`

Default label mode. Each vertex displays the dimension vector of the corresponding indecomposable module. The source is the `digraph Quiver` block in `filename.log`.

`show label`

Replaces each dimension vector with the internal node number. Useful when comparing the canvas with `filename.log`, where all tables reference modules by node number.

`show custom label`

Displays user-defined labels when available; falls back to dimension-vector labels otherwise.

> **Tip**
>
> Double-click a vertex to **customise its label**. The input supports basic Unicode. For instance, `N^2` renders as `N²`.

`Ctrl+L hide/show UI`

Hides or restores the user interface. Useful for screenshots or for arranging a large graph.

`Day/Night`

Toggles between day mode and night mode. Night mode keeps mathematical colours stable where possible, but changes the main canvas, controls, drawers, and floating windows to black backgrounds with white text. AR irreducible arrows are black in day mode and white in night mode; the colour legend updates this convention dynamically.

Sidebar: `View`

`PD`

Displays the projective dimension of each indecomposable module as a floating label; `-1` denotes ∞ or an unresolved value. Data source:

```
PDID := [[node, pd, id], ...];
```

`ID`

Displays the injective dimension of each module, using the third entry of the same `PDID` table.

`Top`

Displays the simple factors of $\mathrm{top}(M)$ for each indecomposable module M, with multiplicities. Data source:

```
TopSoc := [[node, top, soc], ...];
```

`Soc`

Displays the simple factors of $\mathrm{soc}(M)$ for each indecomposable module M, using the same `TopSoc` table.

`Borders`

Shows or hides vertex borders. Convention:

- blue: injective;
- red: projective;
- purple: projective-injective;
- grey: other indecomposable.

Data source:

```
Projective modules found (Node IDs): [...]
Injective modules found (Node IDs):  [...]
```

Sidebar: `Quivers`

Overlays auxiliary quivers on the AR-quiver canvas.

`AR irreducible arrows`

Shows or hides the irreducible morphisms of the AR-quiver (black arrows in day mode, white arrows in night mode). Data source:

```
digraph Quiver { ... }
```

Repeated arrows may be collapsed visually; multiplicity information is preserved.

`Translation quiver τ`

Shows the AR translation quiver. A golden arrow $M \longrightarrow N$ indicates $N = \tau M$. Data source:

```
digraph TranslationQuiver { ... }
```

`Syzygy quiver`

Shows the syzygy quiver. A pink arrow $X \longrightarrow Y$ indicates $Y \in \mathrm{add}(\Omega X)$. Data source:

```
digraph SyzygySummand { ... }
```

The calculator iterates this quiver to compute $\Omega^n(X)$.

`Cosyzygy quiver`

Shows the cosyzygy quiver. A green arrow $X \longrightarrow Y$ indicates $Y \in \mathrm{add}(\Sigma X)$. Data source:

```
digraph CosyzygySummand { ... }
```

The calculator iterates this quiver to compute $\Sigma^n(X)$.

`Radical quiver`

Shows the radical quiver. A cyan arrow $X \longrightarrow Y$ indicates $Y \in \mathrm{add}(\mathrm{Rad}(X))$; repeated arrows encode multiplicities. Data source:

```
digraph RadicalSummand { ... }
```

The calculator iterates this quiver to compute $\mathrm{Rad}^n(X)$.

`Coradical quiver`

Shows the coradical quiver. A purple arrow $X \longrightarrow Y$ indicates $Y \in \mathrm{add}(X/\mathrm{Soc}(X))$; repeated arrows encode multiplicities. Data source:

```
digraph CoradicalSummand { ... }
```

The calculator iterates this quiver to compute $\mathrm{Corad}^n(X)$.

`Hom dimension quiver`

Shows nonzero Hom dimensions. A brown arrow $M \longrightarrow N$ with label `k` (unlabelled means `k=1`) indicates $\dim \mathrm{Hom}(M, N) = k$. Data source:

```
digraph HomDim { ... }
```

`Ext dimension quiver`

Shows nonzero Ext^1 dimensions. A red arrow $M \longrightarrow N$ with label `k` (unlabelled means `k=1`) indicates $\dim \mathrm{Ext}^1(M, N) = k$. Data source:

```
digraph ExtDim { ... }
```

Combined with the syzygy quiver, this data is used by the calculator to evaluate $\mathrm{Ext}^{\geq 1}$ dimensions.

Extension middle-term table

The computation log also contains an extension middle-term table:

```
# --- ExtensionMiddleTermTable --- #
ExtensionMiddleTermTable := [
  [rec(sub := i, quot := j, mids := [[...], ...]), ...],
  ...
];;
```

The entry with `sub := i` and `quot := j` records known middle terms of short exact sequences

$$0 \to M_i \to E \to M_j \to 0.$$

Each row of `mids` is the list of indecomposable summands of one possible middle term. For example, `mids := [[2,3], [1,12]]` means that the known middle terms include both $M_2 \oplus M_3$ and $M_1 \oplus M_{12}$.

By default this table is a fast **known-middle-term** table, not a full enumeration of all extension classes. It includes:

- split extensions;
- the fact that pairs with $\dim \mathrm{Ext}^1(M_j, M_i) = 0$ have only the split middle term;
- non-split sequences coming from radical/top, socle/coradical, projective covers/syzygies, and injective envelopes/cosyzygies;
- middle terms inferred from AR translation meshes and dimension-compatible commutative squares in the AR-quiver;
- additional middle terms obtained by iteratively composing such pushout/pullback squares, using the fact that a composite of pushout/pullback squares is again a pushout/pullback square. This iteration is guarded by `max_extension_square_composition_iterations` and `max_extension_square_composition_rectangles` to avoid blow-up on large examples.

To request the slower full pushout-based table, set the GAP variable

```
compute_full_extension_table := true;;
```

before running the computation. Over an infinite field, the full mode still uses the split extension plus basis representatives produced by `ExtOverAlgebra`; over a finite field it can enumerate finite linear combinations.

`Original quiver Q`

Opens a draggable window displaying the original quiver and its relations. Data source:

```
digraph Q { ... }
rel := ...;
```

The `see filename.txt` button opens the original input. The `Open in q.uiver` button opens the URL stored in the first line of the input file.

The window now uses the same SVG-style quiver rendering as the module-matrix viewer: circular vertices, directed arrows, labels, and a layout derived from the input quiver structure. Vertex and arrow labels keep a halo for readability; in night mode the quiver switches to a black background with white strokes/text.

Sidebar: `Modules`

These buttons highlight special classes of indecomposable modules without altering the graph.

> **Note**
>
> Each property below is closed under direct summands, so it suffices to state it for indecomposable modules.

`Torsionless`

Highlights **non-projective** torsionless modules (cyan). A module M is torsionless if the canonical map

$$M \to D^2M, \quad m \mapsto (f \mapsto f(m))$$

is injective, or equivalently if M embeds into a projective module. Data source:

```
Torsionless modules found (Node IDs): [...]
```

`Reflexive`

Highlights **non-projective** reflexive modules (purple). A module M is reflexive if the canonical map

$$M \to D^2M, \quad m \mapsto (f \mapsto f(m))$$

is an isomorphism. Data source:

```
Reflexive modules found (Node IDs): [...]
```

`Gorenstein projective`

Highlights **non-projective** Gorenstein projective modules (green). Data source:

```
Gorenstein projective modules found (Node IDs): [...]
```

`Gorenstein injective`

Highlights **non-injective** Gorenstein injective modules (red). Data source:

```
Gorenstein injective modules found (Node IDs): [...]
```

Sidebar: `Classes`

Class and tilting lists open in a resizable side drawer. Clicking the same list button again closes the drawer. Rows are sortable where applicable; selecting a row highlights the corresponding modules on the AR-quiver. The drawer path and checkbox path both render the same list content.

`Torsion classes`

Opens a list of torsion pairs. Each row has the form

```
T := [...] | F := [...] | tilting/non-tilting
```

Colour convention: red vertices denote the torsion class $\mathcal{T}$; green vertices denote the torsion-free class $\mathcal{F}$. For a tilting module L, the induced torsion pair satisfies

$$\mathcal{T} = \mathrm{gen}(L) = \mathrm{Ker}\,\mathrm{Ext}^1(L, -), \qquad \mathcal{F} = \mathrm{Ker}\,\mathrm{Hom}(L, -).$$

Data source:

```
# --- TorsionPairTable --- #
T := [...] | F := [...]
```

The `tilting` flag records whether the torsion pair arises from a tilting module.

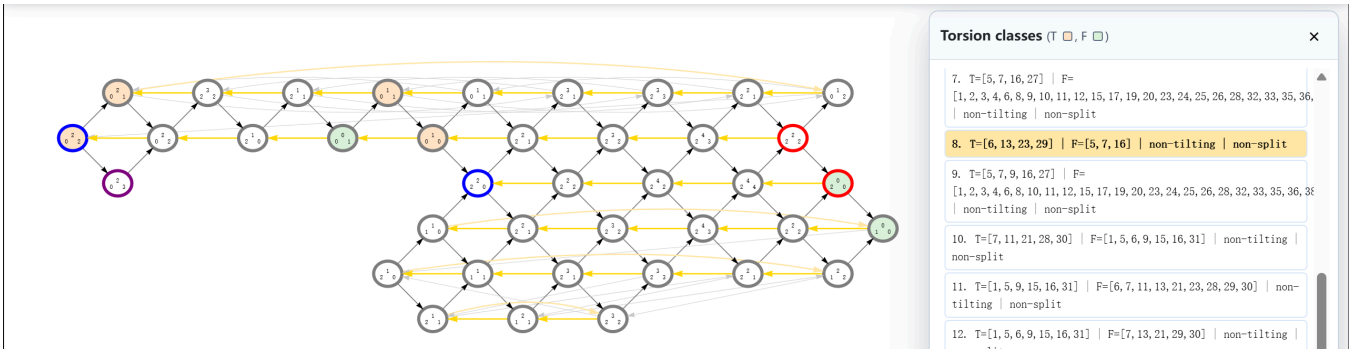
> **Caution**
>
> Distinct tilting modules may induce the same torsion pair.

The `split` flag records whether the torsion pair $(\mathcal{T}, \mathcal{F})$ splits, i.e., $\mathcal{T} \cup \mathcal{F}$ exhausts all indecomposable modules.

`Controls > GAP codes > Extension closure` provides copyable GAP code for computing extension closures from a middle-term table. This is kept out of the generated HTML sidebar list because enumerating all extension-closed classes can be exponentially large. The GAP snippet includes:

- `ExtensionClosureFromTable(X, table)`;
- `IsExtensionClosedClassFromTable(X, table)`;
- `AllExtensionClosedClassesFromTable(table)` for small enough examples.

The HTML still stores `ExtensionMiddleTermTable`, so the calculator can compute the closure of a selected class on demand without rendering the full list.



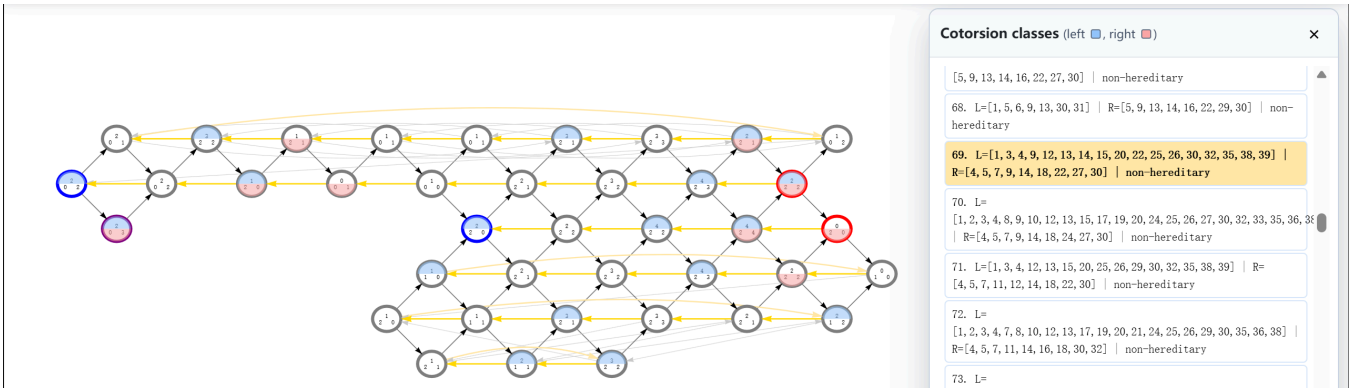
`Cotorsion classes`

Opens a list of cotorsion pairs. Each row has the form

```
L := [...] | R := [...] | Hereditary := true/false
```

Colour convention: blue vertices denote the left class $\mathcal{L}$; red vertices denote the right class $\mathcal{R}$; modules in both classes are half-coloured. The `hereditary` flag records whether $\mathrm{Ext}^{\geq 1}(\mathcal{L}, \mathcal{R}) = 0$. Data source:

```
# --- CotorsionPairTable --- #
L := [...] | R := [...] | Hereditary := ...
```



Sidebar: `Tilting`

`Tilting modules`

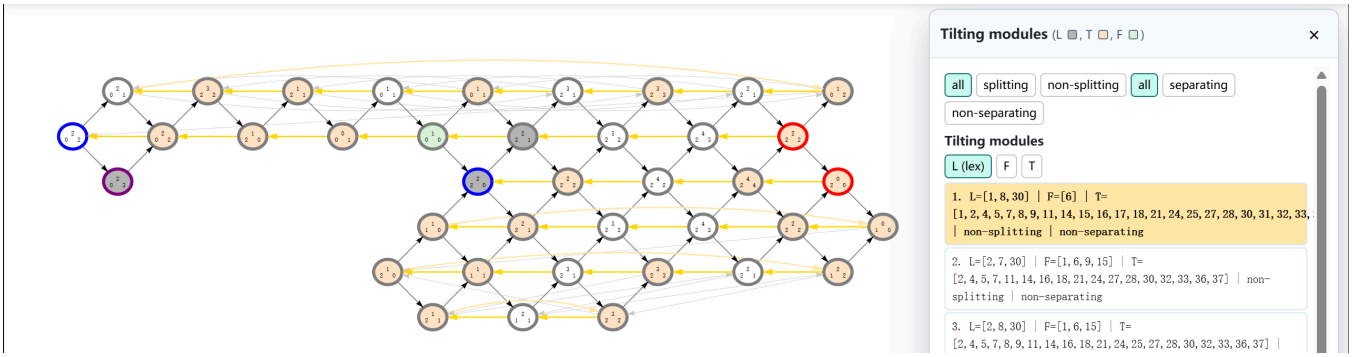
Opens the list of classical tilting modules. A module T is (classical 1-) tilting if $\mathrm{pd}\, T \leq 1$, $\mathrm{Ext}^{\geq 1}(T, T) = 0$, and there exists a short exact sequence $A \rightarrowtail T^0 \twoheadrightarrow T^1$ with $T^0, T^1 \in \mathrm{add}(T)$.

Colour convention: grey vertices are the indecomposable summands of the selected tilting module; red vertices are in the induced torsion class; green vertices are in the induced torsion-free class. Data source:

```
L := [...]
F := [...]
T := [...]
```

where `L` is the tilting module, `T` is the torsion class, and `F` is the torsion-free class.

The `splitting` tag records whether every indecomposable module in `F` has injective dimension at most one. The `separating` tag records whether the induced torsion pair is split.



`Support τ-tilting`

Opens the list of support τ-tilting pairs. A pair (P, M) is support τ-tilting if P is projective, M is τ-rigid, $\mathrm{Hom}(P, M) = 0$, and $|\mathrm{ind}\, P| + |\mathrm{ind}\, M|$ equals the number of vertices of the original quiver. Data source:

```
# --- SupportTauTiltingTable --- #
P := [...] | M := [...]
```

Colour convention: light blue vertices denote the projective part P; grey vertices denote the indecomposable summands of M. The list is rendered in the side drawer and supports sorting by `P` or `M`.

`Almost support τ-tilting`

Opens the list of almost support τ-tilting pairs, which have the same format as support τ-tilting pairs but with total summand count one less than the number of vertices. Data source:

```
# --- AlmostSupportTauTiltingTable --- #
P := [...] | M := [...]
```

The same colour convention is used: light blue for P and grey for M.

Sidebar: `GAP codes`

This section contains copyable GAP/QPA snippets derived from the current input. They are intended for experiments that are better run in GAP than in the browser.

- `Generate this quiver`: reconstructs the quiver, relations, algebra, projective modules, injective modules, simples, and serialised indecomposables.
- `Find gen(-)`: code for computing generated classes.
- `Find cogen(-)`: code for computing cogenerated classes.
- `Extension closure`: code for computing extension closures from the middle-term table.
- `Ext basis sequences`: code for extracting representative short exact sequences from an Ext^1 basis.
- `Hom basis matrices`: code for inspecting bases of Hom spaces as matrices.
- `Module matrices`: code for printing the linear maps defining selected modules.

Sidebar: `Tools`

Tool windows are draggable and resizable where appropriate. For `Matrices`, `Class inspector`, `Module matrix`, `Traverse`, `Calculator`, `Export AR-quiver to TeX`, `Display code`, and `Color legend`, clicking the same control button again closes the corresponding window and clears its active state.

`Matrices`

Displays adjacency-style matrices for several quiver relations:

- Hom dimensions;
- Ext^1 dimensions;
- τ translation;

- syzygy and cosyzygy summands;
- radical and coradical summands.

The output can be shown as plain text, Sage syntax, or LaTeX `pmatrix` syntax. The first line records the node-number order used for rows and columns.

`Class inspector`

Inspects a class of modules entered as a list of node numbers. It reports its size, whether it appears as a torsion/torsion-free or cotorsion-side class, and whether it is closed under syzygy, cosyzygy, radical, or coradical operations. Buttons allow using the selected vertices, highlighting the class, storing one class, and comparing the stored class with the current class.

`Module matrix`

Displays the explicit linear maps of a selected indecomposable module. The graphical view uses the original quiver layout and can show either dimensions plus matrices or a simpler node/edge view. `Use selected` fills the module label from the selected AR-quiver vertex; `Reset layout` restores the module-matrix node positions.

`Traverse`

Iterates a class through one of the available quiver operations: τ, τ^{-1}, syzygy, cosyzygy, radical, or coradical. Enter a seed list and a number of steps, or use the currently selected vertices as the seed.

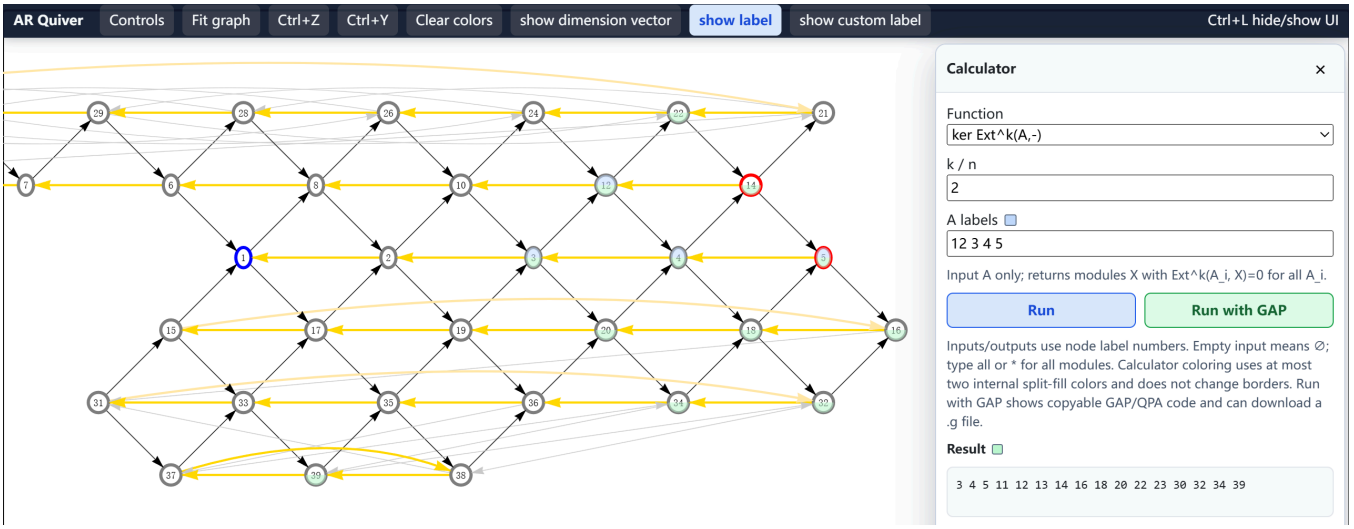
`Calculator`

Performs operations on data stored in the HTML file. Inputs and outputs use node numbers.

Available operations:

- `dim Ext^k(A,B)`: computes $\sum_{i,j} \dim \mathrm{Ext}^k(A_i, B_j)$ over all selected labels.
- `extension closure(A)`: computes the smallest class containing the selected labels and closed under the known middle terms in `ExtensionMiddleTermTable`.
- `middle terms A→?→B`: requires one indecomposable label in `A` and one in `B`, and lists the known middle terms of short exact sequences $0 \to A \to E \to B \to 0$, one middle term per line.
- `ker Ext^k(A,-)`: returns all X with $\mathrm{Ext}^k(A_i, X) = 0$ for every selected A_i.
- `ker Ext^k(-,B)`: returns all X with $\mathrm{Ext}^k(X, B_j) = 0$ for every selected B_j.
- `Ω^n(A)`: iterates the syzygy quiver n times.
- `Σ^n(A)`: iterates the cosyzygy quiver n times.
- `Rad^n(A)`: iterates the radical quiver n times.
- `Corad^n(A)`: iterates the coradical quiver n times.

Notation: `3 + 5` denotes $M_3 \oplus M_5$; `3^2 + 5` denotes $M_3^{\oplus 2} \oplus M_5$. Empty output is displayed as `∅`.



`Generate this quiver` / `Run with GAP`

The `GAP codes` section provides copyable scripts, and the calculator also has a `Run with GAP` helper. These snippets reconstruct the quiver, algebra, projective modules, injective modules, simple modules, and serialised indecomposable modules when needed. Generated scripts can be downloaded as `.g` files from the corresponding code panel.

`Export AR-quiver to TeX`

Exports the current AR-quiver layout as copyable TEX code. Two modes are available:

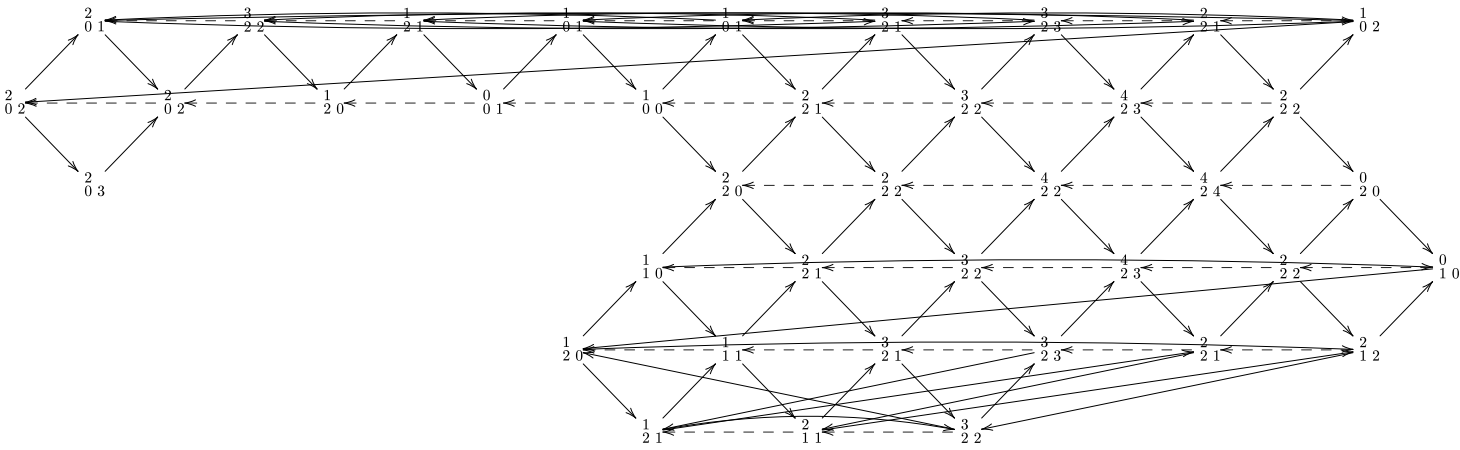
- `xymatrix`: matrix-style output for xy-pic.
- `tikz`: `tikzpicture` output preserving vertex positions, dashed τ-arrows, and curved arrows via `bend left` / `bend right`.

Required preamble packages:

```
\usepackage[all]{xy}       % for xymatrix
\usepackage{tikz,amsmath} % for tikz
```

> **Caution**
>
> Use `tikz` when the picture contains **curved dashed arrows**, as `xymatrix` has limited support for that combination.

`Typora` supports `xymatrix`.



`Display code`

Records the current node positions, arrow-curve offsets, and dimmed-arrow state as copyable code. The layout can be restored later or transferred to another `html` file with the same underlying graph. Mathematical data are not included. The button acts as a toggle: click it again to close the display-code window.

For instance, a display code has the form

```
ARQ3.<node-position-code>.<curve-code>.<dimmed-arrow-code>
```

`Color legend`

Opens a draggable legend summarising all visual conventions: vertex borders, module-class colours, AR arrows, τ-arrows, syzygy, cosyzygy, radical, coradical, Hom and Ext^1 arrows, torsion/cotorsion colours, tilting colours, support τ-tilting colours, calculator colours, and floating labels. The legend follows day/night mode; in particular, the `AR irreducible arrow` swatch is black in day mode and white in night mode.

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